Research Brief

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Working Title

Who Benefits from Flood-Mitigation Investment? Housing Capitalization, Spatial Spillovers, and Distributional Effects in Flood-Prone Communities

Broader Research Agenda: Place-Based Public Investment, Housing Markets, and Spatial Inequality

Research Motivation

My research examines how place-based public investments reshape housing and land markets, and how the resulting market responses redistribute wealth, risk, and the ability to remain in place. This agenda grew out of fifteen years of working in urban development and renewal in China, including new-town and industrial-park development, urban regeneration, and waterfront projects. I observed how new infrastructure repositions surrounding land for higher-value use, changes who can afford to stay, and, through land acquisition and resettlement, distributes benefits and burdens unevenly among residents.

These experiences led me to a broader question: when a spatially targeted intervention changes environmental risk, accessibility, amenities, or expectations, how are those changes transmitted through housing and land markets, and who ultimately gains or loses? A project can meet its stated objective in aggregate yet produce sharply unequal consequences, such as capital gains for incumbent homeowners, affordability pressures for renters and prospective buyers, shifts in housing supply and neighborhood composition, or benefits concentrated in already advantaged places. My interest is not only in whether place-based investment works, but in how its effects are capitalized, distributed, and mediated by local market and institutional conditions.

In my doctoral research, I examine flood-mitigation investment as a primary empirical setting because it combines spatially targeted public spending, measurable risk reduction, and consequential housing-market effects.

Research Questions

1. What is the causal effect of flood-mitigation project completion on nearby housing values, and how do treatment effects evolve before and after completion?
2. How do these effects vary with distance from the protected area and across flood-risk boundaries, and to what extent do they spill over into nearby untreated housing markets?
3. How are capitalization effects distributed across neighborhoods with different income levels, social vulnerability, and tenure composition, and what do these differences imply for housing wealth, affordability, and neighborhood change?

Proposed Empirical Strategy

The project would link FEMA mitigation-project records with parcel-level housing transactions, flood-risk geography, neighborhood characteristics, and project timelines. The primary design would use difference-in-differences and event-study methods suited to staggered treatment timing and heterogeneous effects, estimating how housing values change around project completion. To identify spatial spillovers and avoid contaminating comparison areas, the analysis would model treatment exposure across distance bands around protected areas and test alternative buffer exclusions for nearby untreated properties. Where defensible geographic boundaries and covariate continuity support the design, spatial regression-discontinuity analyses would provide complementary local evidence. Because projects are not randomly sited, the analysis would account for pre-treatment flood exposure, property-market conditions, institutional capacity, and local fiscal conditions, and test robustness across alternative comparison groups. Distributional heterogeneity would be examined across income, social vulnerability, tenure composition, and baseline market conditions, with interpretable machine learning used selectively as a supplement to causal identification. Where data permit, additional outcomes would include rents, residential turnover, building activity, and housing-supply responses.

Research Preparation

My writing sample, "Hazard or Amenity?," provides the empirical foundation for this agenda through a hedonic study of 47,763 single-family sales in Pierce County, Washington, linking parcel-level transactions with FEMA flood-risk, waterway, and neighborhood data. The study shows that estimates of flood-risk capitalization must account for waterfront amenities and neighborhood context. It also exposes the limits of cross-sectional analysis, underscoring the need for causal identification of how mitigation investment affects housing values, generates spatial spillovers, and distributes benefits across communities.

Expected Contributions

The project advances research on place-based public investment by supplying causal evidence on how mitigation investment is transmitted through housing markets and across space, extending beyond conventional estimates of flood-risk capitalization. For housing and climate-adaptation policy, it examines when public investment generates housing-wealth gains, affordability pressures, or neighborhood change, and whether these market benefits accrue to socially vulnerable communities or disproportionately to advantaged owners and neighborhoods. The project's methodological contribution is an urban-informatics framework that integrates causal inference, spatial exposure modeling, and heterogeneous-effects analysis, transferable to other evaluations of the spatial and distributional consequences of place-based investment. Flood mitigation is the primary doctoral application of this broader agenda, which can later inform the evaluation of other spatially targeted interventions, such as transportation, green infrastructure, and environmental remediation, that reshape housing and land markets.